

Air Pencil — Setup & Run Commands

Quick reference to set up and run the project on any Windows laptop.

0. Before you start (one-time per laptop)

MediaPipe (used for hand tracking) does not support the newest Python versions yet. Python **3.11** must be installed on the laptop, even if a newer Python version is already present. Both can exist side by side without any conflict.

If Python is not installed at all, download the Python Install Manager from python.org, then run:

```
py install 3.11
```

Note: if you see a warning about the 'legacy py.exe command', uninstall the old **Python Launcher** app from Installed Apps first, then retry.

1. Open the project folder in a terminal

In VS Code: File > Open Folder → select the folder containing `air_pencil.py` and `requirements.txt`. Then open Terminal > New Terminal.

2. Create a virtual environment (Python 3.11)

```
py -3.11 -m venv venv
```

This creates an isolated 'venv' folder inside your project, using Python 3.11 only.

3. Activate the virtual environment

```
venv\Scripts\activate
```

You should see (venv) appear at the start of the terminal line. If you get a security/execution-policy error instead, run the command below once, type Y, then retry:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

4. Install the required libraries

```
pip install opencv-python numpy mediapipe==0.10.21
```

This step needs an internet connection and may take a few minutes (mediapipe is a large package).

5. Run the project

```
python air_pencil.py
```

A webcam window will open. Allow camera access if prompted.

6. Every time you reopen the project later

You do NOT need to repeat steps 2 and 4 again on the same laptop. Just do:

```
venv\Scripts\activate  
python air_pencil.py
```

Gesture Controls (once the app is running)

Gesture	Action
Only index finger up	Draw / erase (pen down)
Index + middle finger up, over top color bar	Select a color or the eraser
Close hand into a fist	Pen up — lift and reposition without drawing
Open palm (all 5 fingers up)	Clear the entire canvas
Keyboard: s	Save current drawing as PNG (saved_drawings/ folder)
Keyboard: q or Esc	Quit the application

Tip: to write letter by letter, close your fist between letters to lift the pen, move your hand, then open your index finger again to continue.